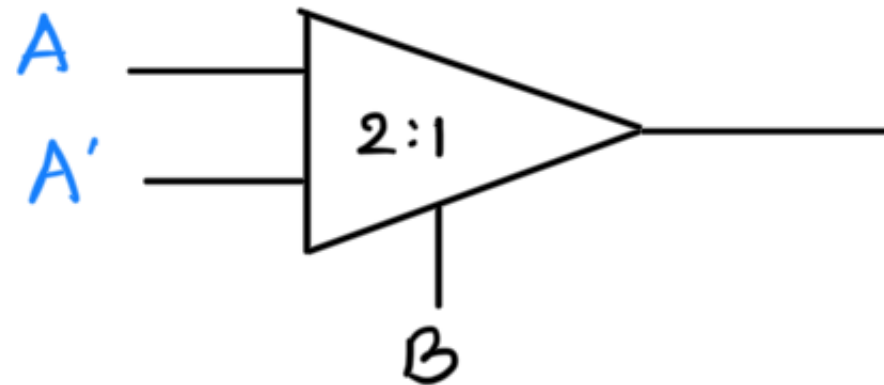


Quiz - 3

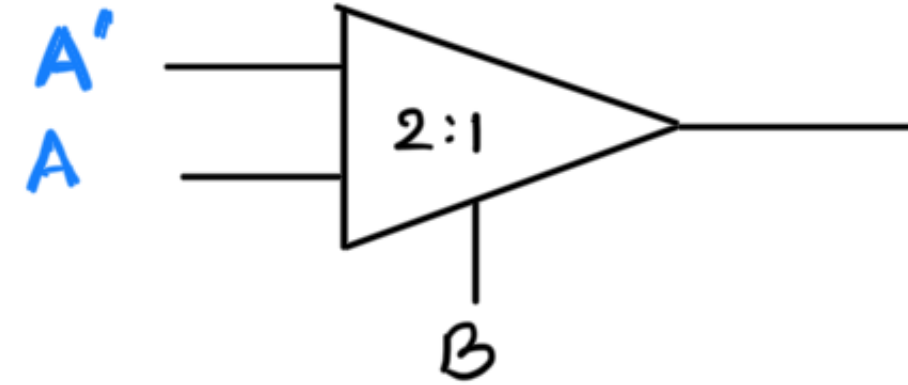
① XOR using 2:1 mux



A	B	XOR
0	0	0
0	1	1
1	0	1
1	1	0

	I_0	I_1
A'	0	1
A	2	3
	A	A'

① XNOR using 2:1 mux



A	B	XNOR
0	0	1
0	1	0
1	0	0
1	1	1

	I_0	I_1
A'	0	1
A	2	3
	A'	A

①①

I_3	I_2	I_1	I_0	A_1	A_0
0	0	0	0	X	X
0	0	0	1	0	0
0	0	1	0	0	1
0	0	1	1	0	1

A_1

$I_3 I_2$ \ $I_1 I_0$	00	01	11	10
00	X			
01	1	1		1

0	1	0	0	1	0
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	1	0	1
1	0	0	0	1	1
1	0	0	1	1	1
1	0	1	0	1	1
1	0	1	1	0	1
1	1	0	0	1	1
1	1	0	1	1	0
1	1	1	0	1	0
1	1	1	1	0	0

$$= l_2 l_1' + l_3 l_1' + l_2 l_0' + l_3 l_0'$$

A₀

$l_3 l_2 \backslash l_1 l_0$

	00	01	11	10
00	X		1	1
01			1	
11	1			
10	1	1	1	1

$$= l_3 l_2' + l_2' l_1 + l_3 l_1' l_0' + l_3' l_1 l_0$$